Benchmarking Confidence Calibration in Spatial Transcriptomics Models Under Distribution Shift

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Abstract

We present a preliminary, fully reproducible study of confidence calibration in a spatial transcriptomics deconvolution model (DestVI) under a controlled deployment-time shift. Unlike a protocol that retrains a model on each shifted distribution in turn, we train and calibrate a single DestVI model exclusively on held-out clean data, then evaluate its frozen predictions and fixed post-hoc calibration mappings as capture efficiency is progressively reduced. We find that uncalibrated confidence exhibits a modest but consistent, dose-dependent increase in Expected Calibration Error (ECE) that reaches strong statistical significance at every shift magnitude tested (all $p < 0.001$), consistent with prior evidence that calibration degrades under distribution shift. We further show that Isotonic Regression, fit on clean calibration data, retains a numerical ECE advantage over Temperature Scaling at every shift magnitude, but this advantage narrows from roughly 9.5× at baseline to 3.9× under the most severe shift tested, with its variance growing nearly three-fold over the same range, while Temperature Scaling remains comparatively stable throughout.

Keywords: Spatial Transcriptomics, Confidence Calibration, Variational Inference, Distribution Shift

Data and Code Availability We utilize the public Mouse Cortex scRNA-seq dataset available via the `squidpy` python package (https://squidpy.readthedocs.io). All code to reproduce this pipeline and data preprocessing is available in our GitHub repository at https://anonymous.4open.science/r/transcriptomics.

Institutional Review Board (IRB) This research uses entirely publicly available, anonymized animal data and does not require IRB approval.

1. Introduction

Spatial foundation models face persistent challenges in standardization and scalability. A critical failure mode is miscalibrated confidence in clinical deconvolution pipelines. Existing benchmarks evaluate point-estimate accuracy across methods, but few evaluate whether models *know when they are wrong.*

Crucially, downstream biological and clinical discovery is sensitive to these miscalibrations. For instance, in tumor microenvironment characterization, if a spatial model confidently but incorrectly predicts an immunosuppressive cell type as the dominant population in a tumor region, it could lead to improper patient stratification for immunotherapies. Ensuring models are correctly calibrated under technical shifts is necessary before these tools can inform high-stakes diagnostic or therapeutic decisions.

In this paper, we focus on the calibration of spatial deconvolution models, specifically DestVI (Lopez et al., 2022), under a controlled deployment-time shift: a single model, trained and calibrated exclusively on clean data, is evaluated without any retraining on progressively shifted held-out test data mimicking capture-rate degradation. This work represents a first, deliberately narrow-scope study intended to establish a rigorous baseline—featuring a fully reproducible pipeline, a genuine train/calibration/test split held constant under shift, and paired significance testing—before extending to real cross-platform shifts and additional model architectures.

2. Related Work

Confidence calibration of modern neural networks was formalized by Guo et al. (2017), demonstrating that deep models often exhibit severe overconfidence but can be realigned using post-hoc methods like temperature scaling or isotonic regression. Subsequent work by Ovadia et al. (2019) established that model calibration frequently degrades under dataset shifts. We extend these foundational frameworks to spatial transcriptomics, where such rigorous evaluations are scarce despite widespread distribution shifts between platforms.

3. Methods

We benchmarked the variational inference model DestVI on mouse cortex pseudo-spots. We evaluated calibration using Expected Calibration Error (ECE). Note that we define our ECE based on the discrete classification task of predicting the dominant (highest proportion) cell type within a spot, utilizing the `netcal` package.

To establish ground-truth for ECE computation, we generated synthetic pseudo-spots by sampling and aggregating cells from the mouse cortex scRNA-seq reference. For each random seed, we partitioned the pseudo-spots into three disjoint, held-out splits: 50% training, 25% calibration, and 25% test. A single DestVI model, utilizing a shared CondSCVI prior, was trained to convergence (up to 200 epochs via ELBO early stopping) exclusively on the clean (unshifted) training split; this frozen model was never retrained or fine-tuned on shifted data. Temperature Scaling and Isotonic Regression were likewise fit exactly once, using the frozen model's predictions on the clean calibration split. To simulate deployment-time cross-platform shifts (such as differences in spatial mRNA capture rates), we synthetically downsampled the read counts of the held-out test split only, via a binomial dropout process, across a dose-response sweep of dropout magnitudes $\{0\%, 20\%, 40\%, 60\%, 80\%\}$, where 0% denotes the unshifted baseline. At each shift magnitude, we ran out-of-sample inference with the frozen DestVI model and applied the fixed Temperature Scaling and Isotonic Regression mappings, without any re-fitting, to obtain calibrated confidences on the shifted test data. This design ensures that both the base model and the post-hoc calibrators are evaluated on genuinely out-of-distribution data relative to what they were trained or calibrated on, rather than being retrained on each shifted distribution in turn.

We report metrics as mean ± standard deviation across 10 independent replicate seeds, each with an independently resampled train/calibration/test split. We evaluated standard softmax-confidence calibration on the model's point-estimate expected proportions using the Expected Calibration Error (ECE) with $M = 10$ bins:

$$\mathrm{ECE} = \sum_{m=1}^{M} \frac{|B_m|}{N} \left|\mathrm{acc}(B_m) - \mathrm{conf}(B_m)\right| \quad (1)$$

where B_m is the set of predictions falling into bin m, and N is the total number of samples evaluated. We evaluated both Temperature Scaling, which divides the pre-softmax logits z_i by a learned scalar $T > 0$ to compute the new confidence $\hat{p}_i = \max \sigma(z_i/T)$, and Isotonic Regression, a non-parametric method that fits a monotonically increasing step function mapping the uncalibrated top-class confidence to the empirical accuracy on the calibration split. We note that our entire benchmarking pipeline is fully reproducible, utilizing fixed random seeds, deterministic CPU training, and publicly available code.

4. Results

Our results show that, evaluated on a held-out clean test split, DestVI has a baseline (0% dropout) Expected Calibration Error (ECE) of 0.215 ± 0.019. This baseline indicates that out-of-the-box variational models exhibit some degree of miscalibration when predicting dominant cell types, even absent any distribution shift.

Under deployment-time capture efficiency shift, the frozen model's uncalibrated ECE increases modestly but consistently with shift severity, from 0.215 ± 0.019 at baseline to 0.259 ± 0.017 under the most severe (80%) dropout tested (Table 1). This increase is highly statistically significant at all evaluated shift magnitudes (all $p < 0.001$, paired t-test vs. baseline). This dose-dependent degradation is consistent with prior evidence that model calibration degrades under

Table 1: Expected Calibration Error (ECE) across varying magnitudes of deployment-time dropout shift, evaluated with a single frozen DestVI model trained and calibrated exclusively on clean, held-out data. All values are reported as mean ± standard deviation across 10 replicate seeds.

Shift	Uncalibrated	p-value (vs. Base)	Temperature Scaled	Isotonic Regression
0% Dropout (1.0, baseline)	0.215 ± 0.019	–	0.067 ± 0.015	0.007 ± 0.006
20% Dropout (0.8)	0.220 ± 0.020	3.88×10^{-5}	0.067 ± 0.017	0.008 ± 0.007
40% Dropout (0.6)	0.228 ± 0.018	2.63×10^{-6}	0.069 ± 0.015	0.009 ± 0.009
60% Dropout (0.4)	0.238 ± 0.018	3.16×10^{-6}	0.069 ± 0.015	0.010 ± 0.009
80% Dropout (0.2)	0.259 ± 0.017	9.18×10^{-6}	0.070 ± 0.012	0.018 ± 0.016

distribution shift (Ovadia et al., 2019), extending that finding to spatial transcriptomics deconvolution.

We applied Temperature Scaling and Isotonic Regression, both fit exclusively on the clean held-out calibration split, and evaluated their fixed mappings on the shifted test data without any re-fitting. As shown in Table 1, Isotonic Regression retains a numerical ECE advantage over Temperature Scaling at every shift magnitude tested, but this advantage narrows substantially as shift severity increases: from a 9.5× reduction relative to Temperature Scaling at baseline (0.007 ± 0.006 vs. 0.067 ± 0.015) to a 3.9× reduction at the most severe shift tested (0.018 ± 0.016 vs. 0.070 ± 0.012, $p = 7.36 \times 10^{-5}$, paired t-test). As visualized in Figure 1, Isotonic Regression's variance grows nearly three-fold over this same range (std $0.006 \to 0.016$), while Temperature Scaling's ECE and variance remain comparatively stable across the entire sweep. We hypothesize that binomial dropout's preservation of expected relative gene-expression ratios provides partial, but not complete, robustness to DestVI's argmax confidences, consistent with the modest rather than severe degradation observed; and that Isotonic Regression's higher-capacity, non-parametric mapping overfits to the exact confidence-accuracy relationship of the clean calibration split, generalizing less reliably than Temperature Scaling's single global parameter as the deployment distribution moves further from that split.

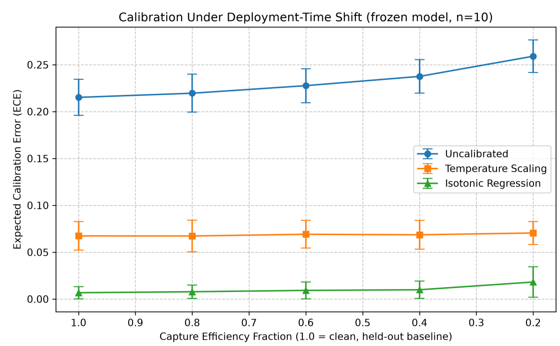

Figure 1: Calibration under deployment-time shift of simulated capture efficiencies, evaluated using a single frozen DestVI model trained and calibrated exclusively on clean data. Error bars denote ±1 standard deviation across 10 random replicate seeds. Isotonic Regression's advantage over Temperature Scaling narrows as shift severity increases.

5. Limitations

We note several limitations in our evaluation. First, to ensure feasibility of multiple runs, models were trained for up to 200 epochs using early stopping (monitoring validation ELBO) on only 50% of available pseudo-spots, with the remainder held out for calibration and test; models trained on more diverse data may exhibit different calibration dynamics. Second, we evaluate robustness using a single synthetic binomial dropout shift on mouse cortex data. To advance this preliminary baseline, future archival work must incorporate real cross-platform datasets—such as the Visium and Slide-seqV2 samples already acquired in our pipeline—and evaluate multiple architectures to validate these findings on authentic biological shifts.

6. Conclusion

Spatial transcriptomics models exhibit a baseline confidence miscalibration that degrades modestly but measurably under deployment-time distribution shift, reaching strong statistical significance at every capture-efficiency reduction tested. Post-hoc recalibration narrows this gap substantially: Isotonic Regression, fit on clean data, retains an advantage over Temperature Scaling at every shift magnitude evaluated, but this advantage shrinks and becomes less reliable as shift severity increases, while Temperature Scaling remains comparatively stable throughout. This suggests that calibrator choice for clinical deployment should account for the expected magnitude of distribution shift, not only in-distribution calibration performance, before trusting high-confidence spatial predictions in clinical contexts.

References

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